1. [Storage system, 15 points] Consider an HDD with the following characteristics:
    - Average seek time = **7**ms
    - Average rotational latency = 3ms
    - Maximum bandwidth = 100MB/s
    - Block size = 4KB

   Suppose an application wants to read 1024 blocks from the HDD.

   1) [11 points] Suppose all the blocks are **sequentially** laid out on the same track.
      a. [1 point] What is the total size of data to be transmitted?

         1024 * 4 kb = 4mb

      b. [3 points] Give an estimate on how long the application needs to wait for the device to start transmitting the data. Explain your answer.

         Seek + Rotation = 7 + 3 = 10 ms

         Rubrics:
         3 points: correct answer and give explanation
         -1 point: if no explanation
         -2 point: if answer is incorrect

      c. [2 points] How much time will the device spend in transmitting the data? Show your derivation.

         4mb / 100 mb/s = 0.04s = 40ms

         Rubrics:
         2 points: both derivation and answer are correct
         1 point: answer is incorrect but give derivation OR answer is correct but no derivation

      d. [1 point] What is the completion time of this workload?

         10 + 40 = 50 ms (seek + rotation + transmitting)

      e. [2 point] Give the formula to compute the actual bandwidth of this workload.

4mb / 0.05 s = 80 ms/s

Rubrics:
0 point if formula is incorrect

f. [2 points] Will the actual bandwidth be the same as the maximum bandwidth? Explain your answer.

No, seek and rotation overhead reduces speed.
Rubrics:
+1 point if answer “No”
+1 point if give correct explanation

2) [4 points] Suppose now all the blocks are **randomly** laid out on the drive.
   a. [2 points] Give the formula to compute the total latency for this random workload.

   1024 * (7 + 3)   Don’t have to calculate it.

   Rubrics:
   0 point if formula is incorrect

   b. [2 points] Will the actual bandwidth for this workload be the same as that for the sequential workload above? Explain your answer.

   No, because we will have 1024 seek/rotation overhead in this case.

   Rubrics:
   +1 point if answer “No”
   +1 point if give correct explanation

2. [Firebase and JSON, 15 points]

   Each of the following curl commands for accessing/updating user information in Firebase has some errors. State what are the errors, why they are errors, and how to fix them.

Suppose user information is stored at:
https://dsci551.firebaseio.com/user.json

Should be https://dsci551.firebaseio.com/users.json ?

1) [3 points] curl -X get ‘https://dsci551.firebaseio.com/users/200’

   Must end with .json and “get” should be upper case

   Rubrics:
   +2 points if find the error and fix it (total 3 points: .json, upper case)
   +1 point if fix them correctly

2) [4 points] curl -X PATCH ‘https://dsci551.firebaseio.com/users/200/age.json’ -d ‘25’
   ‘25’ is not a valid json. It should be 25 if we want it to be stored as number or “25” if we want it to be a string.

   Rubrics:
   +1 points if find one error (total 3 points: .json, $key, 100)
   +1 point if fix them all

3) [4 points] curl -X GET ‘https://dsci551.firebaseio.com/users?orderBy=$key&startAt=200’

   Add .json and $key and 200 needs to be a json string with double quotes.
   +1 points if find the error and fix it (total 4 errors: .json, age, 25, 2)

4) [4 points] curl -X PUT ‘https://dsci551.firebaseio.com/users/200.json’ -d “{‘name’: 25, ‘age’: Null}”

   Null should be lowercase, json strings need to have double quotes.

   Rubrics:
   +1 points if find the error and fix it (total 3 errors: name, age, null)

+1 point if fix them all

3. [XML, 10 points] Consider the following XML file storing the metadata information in the NameNode.

```
<fsimage>
    <inodes>
        <inode>
            <id>100</id>
            <name/>
            <type>directory</type>
        </inode>
        <inode>
            <id>101</id>
            <name>user</name>
            <type>directory</type>
        </inode>
        …
    </inodes>
    <directories>
        <directory>
            <parent>100</parent>
            <child>101</child>
            <child>102</child>
        </directory>
        <directory>
            <parent>101</parent>
            <child>103</child>
        </directory>
        …
    </directories>
</fsimage>
```

Note that: (1) inode type may be either “file” or “directory”; (2) id’s are integers (similar to inode numbers). Example ids are 100, 101, 102, etc.

Write an XPath expression for each of the following questions.

(1) [2 points] Find the name of the directory whose id = **101**. Return the actual name (not element).

/fsimage/inodes/inode[id='101' and type='directory']/name/text()

2 points: correct answer
-1 point: if there is one small error
0 point: if more than two errors

(2) [3 points] Find ids of inodes whose "name" sub-element is NOT empty. Return the actual id value (not element).

/fsimage/inodes/inode[name/text()]/id/text()
Or
/fsimage/inodes/inode[name!='')]/id/text()

3 points: correct answer
-1 point: if there is one small error
0 point: if more than two errors

(3) [2 points] Find the id of the directory which has two children whose ids are 107 and 108. Return the id value (not element).

/fsimage/directories/directory[child='107' and child='108']/parent/text()

2 points: correct answer
-1 point: if there is one small error
0 point: if more than two errors

(4) [3 points] Find names of directories that contain "551". Return names (not elements).

/fsimage/inodes/inode[type='directory' and contains(name, '551')]/name/text()

3 points: correct answer

-1 point: if there is one small error
0 point: if more than two errors

4. [HDFS, 20 points] Suppose an HDFS client wants to implement the following command:

   hdfs dfs **-put** sales.csv /home/dsci551/

   Suppose sales.csv has 256MB, and the replication factor is **3**. Suppose there does not exist a file with the same name under dsci551 directory when the client executes the command.

   1) [2 points] How many blocks are needed to store sales.csv, including all its replicas. Explain your answer.

      2 * 3 blocks, sales.csv need 2 blocks, and three replicas each block

   2) [2 points] Should all replicas of the same block be placed on the same machine? Explain your answer.

      No, will lose them all if on the same machine.

      Rubrics:
      +1 point for "No"
      +1 point for explanation

   3) [2 points] What are the **advantages** of placing two replicas in the same rack?

      Closer to each other, can quickly copy from another replica when one replica is lost.

      Rubrics:
      +2 points for correct advantages

   4) [3 points] What are the **disadvantages** of placing two replicas in the same rack?

      Machines in the same rack share same power supply and cooling, so machines are more likely to fail together, compared to machines in different rack.

      Rubrics:

+3 points for correct disadvantages

5) [4 points] Does the client need to contact NameNode in order to implement the above command? If yes, state what RPC functions will be called and what the purpose of each call is.

    Yes, create() since it is a new file, this will inform namenode to create a new file under dsci551.

    Addblock() this will ask namenode to allocate a new block for the new file.

Rubrics:

+1 points answer 'yes'

+2 points for functions (create, addblock)

+1 point for explanation

6) [2 points] How many nodes will be involved in the pipeline in writing a block from sales.csv? Explain your answer.

    3 datanodes, since replicator factor = 3

Rubrics:

+1 points answering 3

+1 point for explanation

7) [2 points] In the data pipelining process in writing a block synchronous or asynchronous? Explain your answer.

    Asynchronous since it does not wait for the receipt of previous packet before sending next one.

Rubrics:

+1 points answering asynchronous

+1 point for explanation

8) [3 points] What are the advantages and disadvantages of asynchronous compared to synchronous pipelining?

    Advantages: fast to complete, since it is not blocking from sending new packets.

Disadvantages: need to make sure all packets are received (e.g., maintaining a queue, packet in queue when sending, and out of queue when acknowledged).

Rubrics:

+1 point for advantage
+1 point for disadvantage
+1 point for correct explanation of